# 2.3 Focus of a Parabola

Name _____ Period ____ Date _____

Algebra II 2.3 · Student sheet

**I can…**

- identify the focus, directrix, and axis of symmetry of a parabola.
- write equations of parabolas from the focus, directrix, or vertex.
- solve real-life problems involving the focus of a parabola.

**Vocabulary** · fill in the missing word

**Parabola** — the set of all points _____ distant from a point (the focus) and a line (the directrix)

**Focus** — the fixed point inside the _____

**Directrix** — the fixed line outside the _____

**Vertical axis (vertex at origin)** — $y = (1/(4p))x^2$: focus (0, p), _____ $y = -p$

**Horizontal axis (vertex at origin)** — $x = (1/(4p))y^2$: focus (p, 0), _____ $x = -p$

## Part 1 · Equal distances

Focus F(0, 2) and directrix $y = -2$. Check each point.

| Point | Distance to F | Distance to y = −2 | Equal? |
|---|---|---|---|
| (0, 0) | | | |
| (4, 2) | | | |
| (−4, 2) | | | |
| (2, 0.5) | | | |
| (8, 8) | | | |
| (2, 2) | | | |

Plot the points that work and sketch the curve. Equation: _____

## Part 2 · Notes: fill in the blanks

Every point on a parabola is the same distance from the _____ and the _____.

$y = (1/(4p))x^2$: focus _____, directrix _____.

$x = (1/(4p))y^2$: focus _____, directrix _____.

## Part 3 · Examples (work them with the class)

### Example 1

Identify the focus, directrix, and axis of symmetry.

a. $y = -(\frac{1}{12})x^2$ b. $3x^2 - 24y = 0$

### Example 2

Identify the focus, directrix, and axis of symmetry of $x = \frac{1}{4}y^2$. Which way does it open?

### Example 3

Write an equation of the parabola with vertex at the origin and

a. directrix $y = -5$. b. focus $(-2, 0)$.

### Example 4

Write an equation of the parabola with vertex (3, 1) and focus (3, 4). Give the directrix.

### Example 5

A satellite dish has a cross section modeled by $y = (\frac{1}{32})x^2$, with x and y in inches. How far from the vertex should the receiver be placed (at the focus)?

## Part 4 · Partner practice

### Partner 1

Identify the focus and directrix of $y = (\frac{1}{16})x^2$.

**Partner 2**

Write an equation of the parabola with vertex (0, 0) and focus (0, −1).